

CSA5112 - Cryptography and Network Security

2. Monoalphabetic Cipher

Aim: To implement the Monoalphabetic Substitution Cipher using a predefined substitution key.

Algorithm:

1. Start.
2. Read the plaintext.
3. Define a substitution alphabet.
4. Replace each character with its corresponding substitution.
5. Display the ciphertext.
6. Stop.

Code:

```
#include<stdio.h>

int main()
{
    char plain[100], cipher[100];

    char key[]="QWERTYUIOPASDFGHJKLZXCVBNM";

    int i;

    printf("Enter Plain Text: ");

    scanf("%s",plain);

    for(i=0; plain[i]!='\0'; i++)
    {
        cipher[i]=key[plain[i]-'A'];
    }

    cipher[i]='\0';
```

```
printf("Cipher Text: %s",cipher);  
  
return 0;  
  
}
```

Sample Input:

```
Enter Plain Text: HELLO
```

Sample Output:

```
Cipher Text: ITSSG
```

Result:

The Monoalphabetic Cipher was successfully implemented using a substitution key, and the plaintext was encrypted into ciphertext.